

PX4_DXP — Open Bug Register

3WD marking rover · branch Upgrade_Spray @ 7d77565 (Jetson matches) · **701 tests pass**

Compiled 2026-07-22 from six field runs of tes_cross_line; **updated 2026-07-22 post-deploy**.

This markdown is the **source**. OPEN_BUGS.pdf is generated from it —
edit here, then regenerate. The PDF was previously hand-maintained with no source,
which is why it drifted.

Severity is impact on the marking result, not effort.

Verified in-env ≠ **field-proven**. The register says which.

A — CODE LEVEL

#	Bug	Location / parameter	Evidence	Sev	Status
A1	source_file never populates → §8 ABSOLUTE ACCURACY can never run. Two different things named source: the stager writes a filename string , the reader expects a dict .	routes/path.py:1179 wrote "source": result["source"]; bag_autorecord.py:412 read source.get("filepath")	§8 reported unavailable on all 6 bundles; the entire absolute analysis had to be done by hand.	HIGH	■ FIXED e3939e3 · deployed · Jetson-verified: §8 gate RUNS
A2	must_hit lost on every non-staged load. Only the staged path passed it, so vertex provenance never reached those missions.	mission_loading.py:100, routes/mission.py:62, sockets/events.py:119 (vs routes/path.py:1279)	mission_20260722_161014 published must_hit = 0 on a 48-pt path.	HIGH	■ FIXED 21a8b05 · deployed · Jetson: preview 64 pts / 4 must-hit ; matches staged route point-for-point. ■ live /path z=3 unconfirmed
A3	EKF position resets ignored. Origin is fixed, but the vehicle estimate can jump against it mid-mission and nothing notices.	xy_reset_counter, delta_xy — zero references in src/ or server/	Source-verified in ekf_helper.cpp: GNSS resets project through the origin and increment the counter. dd166b3 does not cover this.	HIGH	■ OPEN
A4	outcome == COMPLETE on partial runs. Only a missing recorder_end marked INCOMPLETE; path coverage was never checked.	bag_autorecord.py:550-560	Two runs covered 24/64 and 76/86 waypoints — both reported COMPLETE.	HIGH	■ FIXED 5f820be · deployed · reproduced both bad runs unprompted. ■ PENDING marker needs one real mission
A5	No /spray/session_config subscription. No mode can reach the spray node; dash and point remain schema-only.	spray_controller_node.py:444 (the node's own comment)	Blocks Spray V2 Phase C and D entirely.	HIGH	■ OPEN — top code blocker
A6	Spray plan Rev 3 stale vs shipped code. §5/§7.2/§7.3 describe a speed gate 6523a84 removed, and deny an RPP coupling that now exists.	SPRAY_CONTROLLER_V2_PLAN.md vs spray_controller_node.py:877	Implementing Phase C against it repeats the 16480d9 false-fix pattern.	HIGH	■ OPEN — do before A5
A7	Cross-session origin repeatability. Origin re-derived at each EKF restart from a fresh first fix; needs SET_GPS_GLOBAL_ORIGIN pinning at boot.	not implemented anywhere	One degE7 quantisation step (≤1.11 cm) of offset per session . Within a session, identical.	MED	■ OPEN — see C1 : within-session is now proven, cross-session is not
A8	CPU contention — executor mismatch. MultiThreadedExecutor with MutuallyExclusiveCallbackGroup gives zero parallelism.	ros_node.py executor setup	50–65% of one core. Arm/disarm safety blocks a naive swap.	MED	■ OPEN
A9	must_hit under-marked at entity junctions. A square built from 4 LINE entities flags only 2 of its 4 corners ; independent of corner smoothing.	path_engine/engine.py:1039-1124 (merge step / junction dedup)	Found while verifying A2. Reproduces identically on the staged route, so it predates 21a8b05. If provenance is under-marked, RPP can still simplify real surveyed corners away — 64c12ff would be quietly incomplete.	HIGH	■ NEW, OPEN

Survey tolerance — closed this cycle

SURVEY_TOL_CM = 2.5 was a constant in analyze_mission.py that decided whether §7 FAILS.

Now PathPlanRequest.survey_tolerance_m → staged → manifest → analyser, precedence

--survey-tol-cm > staged > default, with the source printed in every report.

■ 7d77565, deployed. ■ Not yet sent by the mobile frontend.

B — CALIBRATION / HARDWARE (QGC parameters)

None of these are code. All need the rover and a tape measure.

#	Bug	Location / parameter	Evidence	Sev	Status
B1	Dual-antenna heading bias — largest open error. A constant heading error makes pure pursuit settle on a line <i>parallel</i> to the path.	EKF2_GPS_YAW_OFF = 180.0 (a round number — assumed, not measured), GPS_YAW_OFFSET = 180.0	5 of 6 runs drove left ; B spread 3.74 cm, means −0.47 to −2.00 cm. offset $\approx$ lookahead $\times$ sin(err): 2 cm needs only −2.2° at 0.52 m lookahead.	HIGH	■ OPEN — do this first
B2	Differential wheel asymmetry. One counts-per-rev value for two wheels that may not match; same one-sided signature as B1.	RBCLW_COUNTS_REV = 148000, RD_WHEEL_TRACK = 0.470	Indistinguishable from B1 in the data. The cheap separator : drive the same line forward then backward — a heading offset flips sign, a wheel-scale error does not.	MED	■ OPEN
B3	GNSS antenna lever arm. Antenna off the vehicle centreline gives a fixed lateral offset; roll amplifies it.	EKF2_GPS_POS_X/Y/Z = 0 / 0 / −0.4 — Y=0 asserts it is centred	Reference rig: 1.934 m pole, 1.7° tilt = 5.7 cm lateral if uncompensated.	MED	■ OPEN
B4	Nozzle offset uncalibrated. Every measurement to date describes where the antenna went, not where paint went.	nozzle_forward_offset_m, nozzle_lateral_offset_m — declared, unmeasured	Blocks error budget 4. ■ SPRAY_NOZZLE_OFFSET_PLAN.md §3.6 xtrack-gate fix is mandatory before any non-zero offset .	HIGH	■ OPEN
B5	Physical re-survey never done. No log can answer where the paint landed.	process gap — Emlid RS3	Survey CSV shows SampLes=1, 1.7 cm single-epoch RMS. Use averaging (10–30 s/point) when re-surveying .	HIGH	■ OPEN

C — SHIPPED BUT UNVERIFIED (code-proven, field evidence pending)

#	Item	Evidence	Sev	Status
C1	Placement determinism (dd166b3)	Two loads of one staged mission on the Jetson, 2026-07-22: max point-to-point difference 0.0000 cm across 40 sampled of 64 waypoints. Was 0.39–1.53 cm before the fix.	HIGH	■ CLOSED on hardware (within-session only — cross-session is A7)
C2	Survey CSV ingest (211244b)	unit tests only; never run in the field — all missions to date were DXF	MED	■ open
C3	POINT-layer control points	synthetic road case only; never exercised on a dense real drawing	MED	■ open
C4	§8 absolute accuracy auto-runs	was blocked by A1 — unblocked ; gate verified RUNS on the Jetson, but has not yet auto-run on a freshly recorded bundle	HIGH	■ open (needs one drive)
C5	segment→smooth ~5 cm seam	visual only; may already be fixed by 64c12ff. Check /rpp/debug[0] vs [40] at the profile flip.	MED	■ open
C6	Spray pivot-state gate (6523a84)	replay-verified 157→53; field-unverified	MED	■ open
C7	Endpoint under-run	rover rests 0.9–2.0 cm short every run; overshoot never above +0.4 cm	MED	■ measured, no fix attempted
C8	Per-run wander	residual RMS 1.1–1.8 cm between runs after removing the mean offset	MED	■ measured, untouched

Recommended order

1. **B1 heading test** — largest error, **no driving needed**, one QGC parameter.
2. ~~A1, A2, A4~~ — ■ done and deployed; the next run batch records clean provenance.
3. ~~C1 two-load determinism~~ — ■ closed, 0.0000 cm.
4. **B4 + B5** — the only route to knowing where paint lands.
5. **A6** → **A5** — unblocks spray modes (write Rev 4 *before* touching Phase C code).
6. **A9** — under-marked junction vertices; may mean 64c12ff is incomplete.

Field-proven this cycle

- **Vertex deletion** (64c12ff) — 64→4 conditioned points, was 64→2.
- **Georef north-scale** (e483d53) — 2.3255 m against a 2.3255 m WGS84 geodesic, residual 0.000 mm.

Tracking is sound at 0.33–0.88 cm per surveyed vertex with extensions.

The remaining error is offset, not tracking — which is why B1–B3 sit above every code item.

Standing caution

Three bugs shipped in the 2026-07-22 cycle because a test's ground truth **mirrored the bug**:

bare-vertex input bypassing densification; haversine using the same wrong radius as the projection; NavSatFix decoded by tests that bypassed the CDR reader. Be sceptical of any test claiming to validate against "truth" — and check a new test **fails** before the fix lands.